

Personal Reflection — New Skills Required in the AI Era

SWE101 — Topic 2

What Surprised Me Most

Before starting this research, I assumed the main skill software engineers would need in the AI era was simply knowing how to use AI tools — installing GitHub Copilot, learning ChatGPT prompts, and moving on. What surprised me was how much deeper the shift goes.

The statistic that stuck with me most was from the Stack Overflow 2025 Developer Survey: more developers **distrust** AI output (46%) than trust it (33%). I expected developers to be mostly positive about AI accuracy, especially since adoption is so high. Instead, the data shows that experienced engineers are actually the most cautious — the ones who have seen the most code are the least willing to trust AI to generate it unsupervised. That tells me something important: real expertise does not get replaced by AI tools, it gets more valuable because of them.

What Concerns Me About AI in Software Engineering

My biggest concern is skill degradation, especially for junior developers and students like me. If AI tools handle most of the actual code writing, there is a real risk that new engineers never develop a deep understanding of what the code is actually doing. A viral blog post from developer Namanyay Goel in early 2025 raised exactly this concern — that junior developers relying on tools like Copilot are missing the foundational knowledge that used to come from struggling through problems manually.

I find this genuinely worrying. If I use AI to generate code I do not fully understand, I might produce working software today and broken software tomorrow, with no ability to diagnose why. The shortcut could become a long-term disadvantage.

What Opportunities I See

At the same time, I see real opportunity. The research showed that prompt engineering alone carries a 56% wage premium over peers without AI skills (PwC, 2025). System design and architecture roles are seeing 27% demand growth against only 4% supply growth. Engineers who develop strong judgment — knowing when to trust AI output and when to question it — will be genuinely difficult to replace.

I also see an opportunity in the combination of technical and communication skills. As AI handles more routine coding, the engineers who can clearly understand a client's problem, translate it into good system design, and then guide AI tools toward the right solution will be the ones who stand out. That is a skill set I can actively build.

Which Skills I Think Will Remain Most Important

Based on my research, I believe the three most durable skills are:

1. **Critical thinking** — the ability to evaluate whether code, architecture, or a solution is actually correct and appropriate, not just functional on the surface.
 2. **System design** — understanding how large systems fit together is something AI cannot fully replicate, and demand for architects is growing fast.
 3. **Adaptability** — with 80% of engineers expected to upskill by 2027 (Gartner), the engineers who build habits of continuous learning will consistently stay relevant regardless of which tools are popular at any given time.
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Did My Opinion Change After Research?

Yes, somewhat. I started thinking this topic was mainly about learning specific new tools. I now think it is more about developing a way of working — one where I stay in control, maintain deep understanding of fundamentals, use AI to accelerate my work rather than replace my thinking, and continuously update my skills.

I also came in slightly worried that AI would make software engineering less interesting. After researching this, I actually feel the opposite. The routine, repetitive parts of coding may be automated. What remains is the more interesting work: designing systems, solving novel problems, making judgment calls, and collaborating with people. That sounds like a better version of the job, not a worse one.

How I Will Personally Adapt

Going forward, I plan to:

- **Keep learning fundamentals** — algorithms, data structures, system design — so I can evaluate AI output rather than just accepting it.
- **Practice prompt engineering deliberately** — treating it as a real skill to develop, not just casual use of chatbots.
- **Build projects** — real, end-to-end projects that force me to make architecture decisions, not just generate code snippets.
- **Stay curious about AI developments** — following industry surveys, reports, and discussions so I am aware of how the field is shifting.

The AI era does not require less from software engineers. It requires different things — and it rewards engineers who take that seriously.